



DEFENSE ADVANCED RESEARCH PROJECTS AGENCY
3701 NORTH FAIRFAX DRIVE
ARLINGTON, VA 22203-1714

PROCUREMENT GUIDANCE

Funding provided via PBAS Transaction Number 1000Z04268
This document provides official guidance pertaining to ARPA Order M139/00
Performer: University of Pennsylvania Contract No.: NEW

TECHNICAL POINTS OF CONTACT

AGENT (AFRL/IF-NY)

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DARPA

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SPECIAL INSTRUCTIONS

DARPA Comptroller Office: (b) (6)

This document provides guidance for FY 2001 for 6 month(s) of a total 24 month effort.

DARPA has determined that work being funded by this document is not classified as Advisory & Assistance Services as defined in DoD Directive 4205.2.

FFRDC: N/A

Source Selection: Full and Open Competition (F&OC)

FOC Type: BAA

FOC Number: 01-24-524

FOC Date: 03/30/2001

Security: Unclassified

The Chief Information Officer (CIO), DARPA, has reviewed the documentation for this action and concluded that it complies with federal and Department of Defense guidance for the acquisition and management of information technology (IT). Under delegated authority from the CIO, Department of Defense, approval for this action is hereby granted for the amount of \$98,600.00 in IT resources per detailed cost list, attached. This approval assumes the performer's contract includes provisions for the compliance with Federal Acquisition Regulations and other appropriate guidance in acquiring and managing these resources, especially regarding competitive selection, Year 2000 Compliance, inventory control, security and resource disposal. The Contracting Officer assumes responsibility for such provisions not so included.

FUNDING PROFILE

Provided FY 2001	Planned/ Revised FY 2002	Planned/ Revised FY	Planned/ Revised FY	Planned/ Revised FY
Amount: \$502,540.00	\$1,422,460.00			
(b) (6)		04/09/2001	(b) (6)	(b) (6)
(b) (6)	(DARPA Business POC)	(Date)	(Phone)	(Email)
(b) (6)		04/10/2001	(b) (6)	(b) (6)
(b) (6)	(Office Director)	(Date)	(Phone)	(Email)
(b) (6)		04/10/2001		
(b) (6)	Comptroller	(Date)		

SPECIAL INSTRUCTIONS (CONTINUED)

Attachments:

Statement Of Work; Selection Evaluation Document; Technical & Cost Proposal; CBD
Announcement; IT GFE Documentation
Detail list of hardware, software and/or support services required with costs.,
Poor-boy letter

Funds expire for obligation purposes on September 30, 2001.

The purpose of this action is to award \$502,540 of FY 2001 funds to The University of Pennsylvania for their effort entitled "Portable Open Source Security Elements (POSSE)" which was selected under BAA 01-24.

Expected Funding Profile

FY 2001 - \$502,540

FY 2002 - \$1,422,460

Total - \$ 1,925,000

Task 5 Option - \$200,000

Total Contract Value - \$2,125,000

Agent Fees were paid on AO L736.

Objective:

The University of Pennsylvania will perform research in the following areas. However, for Task 5, The University of Pennsylvania shall not, in any way, incur commitments for or otherwise commence performance, unless performance is authorized earlier by written direction from the Contracting Officer.

Task 1 - Audit OpenSSL. OpenSSL will be subjected to a security audit and released.

Task 2 - Portable Kernel Interface. A portable kernel interface to OpenBSD for OpenSSL to communicate with kernel-supported hardware cryptography devices.

Task 3 - OpenSSL Crypto Release. OpenSSL will be released with support for symmetric-keyed encryption hardware (such as 3DES) to increase performance.

Task 4 - File System Enhancements. TrustedBSD file system enhancements will be imported into OpenBSD; ACLs will be built on top of the extended attributes from TrustedBSD.

Task 5 (OPTION) - SEBOS Addition to OpenBSD. A secure bootstrap component imported from the Maryland's SEBOS, will be incorporated into OpenBSD.

Task 6 - User-kernel Interface. The /dev/policy device driver will be incorporated into the OpenBSD release, as a user-kernel interface for security policies.

Negotiate SOW Task 5 as an option, yet include in total contract value, expect cost will be approximately \$200,000.00.

DARPA Internal Use Only

Last Changed: 04/10/2001

ARPA Order: M139/00

Package #: IT 4881

FUND TYPE	TYPE OF ACTION	COMPETITIVE	START DATE	END DATE	LINE NUMBER	FISCAL YEAR	PROGRAM ID	STAGE OF FUNDS	AMOUNT
A	Administrative	Yes			1	2002	CHTSE	Due Bill	\$1,422,460.00
A	Contract	Yes	10/01/2000	12/30/2001	2	2001	CHTSE	New Effort	\$502,540.00

RELATIONSHIP TO PAD

This effort supports: PAD 000025 CHATS ; Task 0001 CHATS**PURPOSE OF THIS ACTION/ACCOMPLISHMENTS**

The purpose of this action is to award \$502,540 of FY 2001 funds to The University of Pennsylvania for their effort entitled "Portable Open Source Security Elements (POSSE)" which was selected under BAA 01-24.

Expected Funding Profile

FY 2001 - \$502,540

FY 2002 - \$1,422,460

Total - \$ 1,925,000

Task 5 Option - \$200,000

Total Contract Value - \$2,125,000

Agent Fees were paid on AO L736.

Objective:

The University of Pennsylvania will perform research in the following areas. However, for Task 5, The University of Pennsylvania shall not, in any way, incur commitments for or otherwise commence performance, unless performance is authorized earlier by written direction from the Contracting Officer.

Task 1 - Audit OpenSSL. OpenSSL will be subjected to a security audit and released.

Task 2 - Portable Kernel Interface. A portable kernel interface to OpenBSD for OpenSSL to communicate with kernel-supported hardware cryptography devices.

Task 3 - OpenSSL Crypto Release. OpenSSL will be released with support for symmetric-keyed encryption hardware (such as 3DES) to increase performance.

Task 4 - File System Enhancements. TrustedBSD file system enhancements will be imported into OpenBSD; ACLs will be built on top of the extended attributes from TrustedBSD.

Task 5 (OPTION) - SEBOS Addition to OpenBSD. A secure bootstrap component imported from the Maryland's SEBOS, will be incorporated into OpenBSD.

Task 6 - User-kernel Interface. The /dev/policy device driver will be incorporated into the OpenBSD release, as a user-kernel interface for security policies.

REMARKS

Negotiate SOW Task 5 as an option, yet include in total contract value, expect cost will be approximately \$200,000.00.

Program Title:

Composable High Assurance Trusted Systems

Title of Effort:

POSSE

By initiating the cover sheet of this action, the Program Manager and office Director certify that information technology (IT) resources being acquired through this action, and their costs, are necessary and appropriate for this effort. Further, this action must be approved by the DARPA Chief Information Officer (CIO). CIO Approval requires the attachment of (1) a detailed list of IT hardware, software and/or services being acquired, and (2) a letter from the contractor on corporate letterhead stating they cannot or will not supply the required IT from their own resources and that they will comply with Federal Acquisition Regulations and other appropriate guidance in acquiring and managing these resources, especially regarding competitive selection, Year 2000 compliance, inventory control, security and disposal.



DEFENSE ADVANCED RESEARCH PROJECTS AGENCY
3701 NORTH FAIRFAX DRIVE
ARLINGTON, VA 22203-1714

PROCUREMENT GUIDANCE

Funding provided via PBAS Transaction Number 02000Z02158
This document provides official guidance pertaining to ARPA Order M139/02
Performer: University of Pennsylvania Contract No.: F30602-01-2-0537

TECHNICAL POINTS OF CONTACT

AGENT (AFRL/IF-NY)

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SPECIAL INSTRUCTIONS

DARPA Comptroller Office: (b) (6)
This document provides guidance for FY 2002 for 12 month(s) of a total 12 month effort.
DARPA has determined that work being funded by this document is not classified as
Advisory & Assistance Services as defined in DoD Directive 4205.2.
FFRDC: N/A
Source Selection: N/A
Security: Unclassified

FUNDING PROFILE

Provided FY 2002	Planned/ Revised FY ____	Planned/ Revised FY ____	Planned/ Revised FY ____	Planned/ Revised FY ____
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Amount: \$560,642.00				
(b) (6)		02/04/2002	(b) (6)	(b) (6)
(b) (6)	(DARPA Business POC)	(Date)	(Phone)	(Email)
(b) (6)		02/04/2002	(b) (6)	(b) (6)
(b) (6)	(Office Director)	(Date)	(Phone)	(Email)
(b) (6)		02/22/2002		
(b) (6)	Comptroller	(Date)		

SPECIAL INSTRUCTIONS (CONTINUED)

This action provides \$560,642 in FY 2002 incremental funds to The University of Pennsylvania for their effort entitled "Portable Open Source Security Elements (POSSE)" on contract F30602-01-2-0537.

Planned activities for FY2002 funds:

- 1.) Hardware Crypto Support
 - a. Both symmetric and asymmetric support
 - b. Important for OpenSSL
- 2.) OpenBSD /dev/policy kernel interface for policy rules
 - a. Prototype policy daemon/kernel done.
 - b. Means of importing many SE Linux features.
- 3.) File system attributes
 - a. Persistent meta-data for objects (ACLs, labels, caps.)
 - b. Absorb from TrustedBSD work

Agent fees will be provided on a separate funding document.

The remaining requirement for FY02 funds of \$589,358 will be provided on a later amendment to AO M139.

DARPA Internal Use Only

Last Changed: 02/22/2002

ARPA Order: M139/02

Package #: AT 4146

FUND TYPE	TYPE OF ACTION	COMPET- ITIVE	START DATE	END DATE	LINE NUMBER	FISCAL YEAR	PROGRAM ID	STAGE OF FUNDS	AMOUNT
A	Contract Incremental	Yes	01/01/2002	12/31/2002	1	2002	GHTSE	Increment	\$560,642.00

RELATIONSHIP TO PAD

This effort supports: PAD 000025 CHATS ; Task 0001 CHATS

PURPOSE OF THIS ACTION/ACCOMPLISHMENTS

This action provides \$560,642 in FY 2002 incremental funds to The University of Pennsylvania for their effort entitled "Portable Open Source Security Elements (POSSE),".

Planned activities for FY2002 funds:

- 1.) Hardware Crypto Support
 - a. Both symmetric and asymmetric support
 - b. Important for OpenSSL
- 2.) OpenBSD /dev/policy kernel interface for policy rules
 - a. Prototype policy daemon/kernel done.
 - b. Means of importing many SE Linux features.
- 3.) File system attributes
 - a. Persistent meta-data for objects (ACLs, labels, caps.)
 - b. Absorb from TrustedBSD work

FY01 Accomplishments:

- 1.) OpenBSD-inspired changes in OpenSSL crypto framework
- 2.) OpenSSL hardware crypto support now working on hifn (Hifn7951) card
- 3.) PowerPC support

REMARKS**Reference Information:**

LINE NUMBER	FISCAL YEAR	PROGRAM	REFERENCE DOCUMENT NUMBER	REFERENCED LINE NUMBER	REFERENCED AMOUNT	FINAL FLAG
1	2002	GHTSE	AT 4115	1	\$560,642.00	No

The comptroller office reduced this due bill by \$589,358 from \$1,150,000 to \$560,642 due to 2002 tax input for program GHTSE on AO M139/TX, AT 4115. The \$589,358 needed to fulfill the requirement to 12/02 to be provided on a later amendment to AO M139.

EXPENDITURE DATA:

AO: M139/01

Contractor: UPENN

Contract #: F30602-01-2-0537

Contract Start Date: 6/25/2001

Projected End Date: 6/25/2004

Total Contract Value: \$1,899,730

Total Provided To Date: \$502,540

Date Current \$ Exhausted: 12/29/2001

Expended as of 12/31/01: \$502,540

Unexpended Funds: \$0

Projected Monthly Burn Rate: \$95,833

Requirements through 12/02: \$1,150,000 (\$560,642 is provided on this AO M139/02, the remaining requirement of \$589,358 will follow on another amendment to this AO)

This amendment will carry the performer through MAY 02.

Program Title:

CHATS

Title of Effort:

POSSE



DEFENSE ADVANCED RESEARCH PROJECTS AGENCY
3701 NORTH FAIRFAX DRIVE
ARLINGTON, VA 22203-1714

PROCUREMENT GUIDANCE

Funding provided via PBAS Transaction Number 02000Z03645
This document provides official guidance pertaining to ARPA Order M139/03
Performer: University of Pennsylvania Contract No.: F30602-01-2-0537

TECHNICAL POINTS OF CONTACT

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SPECIAL INSTRUCTIONS

DARPA Comptroller Office: (b) (6)
This document provides guidance for FY 2002 for 12 month(s) of a total 12 month effort.
DARPA has determined that work being funded by this document is not classified as
Advisory & Assistance Services as defined in DoD Directive 4205.2.
FFRDC: N/A
Source Selection: N/A
Security: Unclassified

FUNDING PROFILE

Provided FY 2002	Planned/ Revised FY ____	Planned/ Revised FY ____	Planned/ Revised FY ____	Planned/ Revised FY ____
Amount: \$589,358.00				
(b) (6)		03/06/2002	(b) (6)	(b) (6)
(b) (6)	(DARPA Business POC)	(Date)	(b) (6)	(Phone)
(b) (6)				(Email)
(b) (6)		03/07/2002	(b) (6)	(b) (6)
(b) (6)	(Office Director)	(Date)	(b) (6)	(Phone)
(b) (6)				(Email)
(b) (6)		03/21/2002		
(b) (6)	Comptroller	(Date)		

SPECIAL INSTRUCTIONS (CONTINUED)

This action provides \$589,358 in FY 2002 incremental funds to The University of Pennsylvania for their effort entitled "Portable Open Source Security Elements (POSSE)" on contract F30602-01-2-0537.

Planned activities for FY2002 funds:

- 1.) Hardware Crypto Support
 - a. Both symmetric and asymmetric support
 - b. Important for OpenSSL
- 2.) Open BSD/dev/policy kernel interface for policy rules
 - a. Prototype policy daemon/kernel done.
 - b. Means of importing many SE Linux features.
- 3.) File system attributes
 - a. Persistent meta-data for objects (ACLs, labels, caps.)
 - b. Absorb from TrustedBSD work

Agent fees will be provided on AO L736/01.

DARPA Internal Use Only

Last Changed: 03/21/2002

ARPA Order: M139/03

Package #: AT 4306

FUND TYPE	TYPE OF ACTION	COMPET- ITIVE	START DATE	END DATE	LINE NUMBER	FISCAL YEAR	PROGRAM ID	STAGE OF FUNDS	AMOUNT
A	Contract Incremental No		01/01/2002	12/31/2002	1	2002	GHTSE	New Effort	\$589,358.00

RELATIONSHIP TO PAD

This effort supports: PAD 000025 CHATS ; Task 0001 CHATS

PURPOSE OF THIS ACTION/ACCOMPLISHMENTS

This action provides \$589,358 in FY 2002 incremental funds to The University of Pennsylvania for their effort entitled "Portable Open Source Security Elements (POSSE),".

Planned activities for FY2002 funds:

- 1.) Hardware Crypto Support
 - a. Both symmetric and asymmetric support
 - b. Important for OpenSSL
- 2.) Open BSD/dev/policy kernel interface for policy rules
 - a. Prototype policy daemon/kernel done.
 - b. Means of importing many SE Linux features.
- 3.) File system attributes
 - a. Persistent meta-data for objects (ACLs, labels, caps.)
 - b. Absorb from TrustedBSD work

FY01 Accomplishments:

- 1.) Open BSD inspired changes in Open SSL crypto framework
- 2.) Open SSL hardware crypto support now working on hifn (Hifn7951) card
- 3.) Power PC support

REMARKS

The comptroller office reduced the original due bill by \$589,358 from \$1,150,000 to \$560,642 due to 2002 tax input for program GHTSE on AO M139/TX. Therefore \$589,358 needed to fulfill the requirement to 12/02, is being provided with this AO.

EXPENDITURE DATA:

AO: M139/03

Contractor: UPENN

Contract #: F30602-01-2-0537

Contract Start Date: 6/25/2001

Projected End Date: 6/25/2004

Total Contract Value: \$1,899,730

Total Provided To Date: \$502,540

Date Current \$ Exhausted: 12/29/2001

Expended as of 12/31/01: \$502,540

Unexpended Funds: \$0

Projected Monthly Burn Rate: \$95,833

Requirements through 12/02: \$1,150,000 (\$589,358 is being provided this AO, M139/02 previously provided \$560,642 of FY2002 funds)

Program Title:

CHATS

Title of Effort:

POSSE



DEFENSE ADVANCED RESEARCH PROJECTS AGENCY
3701 NORTH FAIRFAX DRIVE
ARLINGTON, VA 22203-1714

PROCUREMENT GUIDANCE

Funding provided via PBAS Transaction Number 04000Z00489
This document provides official guidance pertaining to ARPA Order M139/07
Performer: University of Pennsylvania Contract No.: F30602-01-2-0537

TECHNICAL POINTS OF CONTACT

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Email: (b) (6)

SPECIAL INSTRUCTIONS

DARPA Comptroller Office: (b) (6)
This document provides guidance for FY 2003 for 12 month(s) of a total 12 month effort.
DARPA has determined that work being funded by this document is not classified as
Advisory & Assistance Services as defined in DoD Directive 4205.2.
FFRDC: N/A
Source Selection: Task Addition to Task/Delivery Order Contract
Security: Unclassified
The Electronic and Information Technology developed, procured, maintained or used as a
result of this action must be accessible to people with disabilities in accordance with
Section 508 of the Rehabilitation Act of 1973.

FUNDING PROFILE

Provided FY 2003	Planned/ Revised FY	Planned/ Revised FY	Planned/ Revised FY	Planned/ Revised FY
Amount: \$123,587.00				
(b) (6)		11/07/2003	(b) (6)	(b) (6)
(b) (6)	(DARPA Business POC)	(Date)	(Phone)	(Email)
(b) (6)		11/07/2003	(b) (6)	(b) (6)
(b) (6)	(Office Director)	(Date)	(Phone)	(Email)
(b) (6)		11/17/2003		
(b) (6)	Comptroller	(Date)		

ARPA Order: M139/07

SPECIAL INSTRUCTIONS (CONTINUED)

This action provides \$123,587 in FY03 funds to the University of Pennsylvania (UPA) as a contract addition for their effort entitled "Portable Open Source Security Elements (POSSE)" under contract number F30602-01-2-0537.

These FY03 funds will be used to cover costs UPENN incurred to cancel support for the OpenBSD hack-fest.

This is the final incremental payment for this effort. The agent is instructed to reduce the TCV to \$1,976,127 and work with the performer to descope the effort, as appropriate.

Agent fees will be provided on another funding vehicle.

Attachments:

Email regarding POSSE & final bill.

DARPA Internal Use Only

Last Changed: 11/07/2003

ARPA Order: M139/07

Package #: AT 6140

FUND TYPE	TYPE OF ACTION	COMPET- ITIVE	START DATE	END DATE	LINE NUMBER	FISCAL YEAR	PROGRAM ID	STAGE OF FUNDS	AMOUNT
A	Contract Add	Yes	07/01/2003	06/25/2004	1	2003	GMGTE	New Effort	\$123,587.00

RELATIONSHIP TO PAD

PURPOSE OF THIS ACTION/ACCOMPLISHMENTS

This action provides \$123,587 in FY03 funds to the University of Pennsylvania (UPA) as a contract addition for their effort entitled "Portable Open Source Security Elements (POSSE)" under contract number F30602-01-2-0537. This funds an overrun on a CHATS program.

These FY03 funds will be used to cover costs UPENN incurred to cancel support for the OpenBSD hack-fest.

This is the final incremental payment for this effort. The agent is instructed to reduce the TCV to \$1,976,127 and work with the performer to descope the effort, as appropriate.

REMARKS

The Director of ATO approved this action on October 22, 2003.

Program Title:
ATO MANAGEMENT

PE: Project: Obj Class:
62301E ST-24 2525

Title of Effort:
POSSE - Final Increment

AFRL-IF-RS-TR-2004-230
Final Technical Report
August 2004



PORTABLE OPEN SOURCE SECURITY ELEMENTS (POSSE)

University of Pennsylvania

Sponsored by
Defense Advanced Research Projects Agency
DARPA Order No. M139

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INFORMATION SUBJECT TO EXPORT CONTROL LAWS

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The views and conclusions contained in this document are those of the authors and should not be interpreted as necessarily representing the official policies, either expressed or implied, of the Defense Advanced Research Projects Agency or the U.S. Government.

**AIR FORCE RESEARCH LABORATORY
INFORMATION DIRECTORATE
ROME RESEARCH SITE
ROME, NEW YORK**

AFRL-IF-RS-TR-2004-230 has been reviewed and is approved for publication.

APPROVED:

(b) (6)

Project Engineer

FOR THE DIRECTOR:

(b) (6)

Information Grid Division
Information Directorate

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REPORT DOCUMENTATION PAGE			Form Approved OMB No. 074-0188	
Public reporting burden for this collection of information is estimated to average 1 hour per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing this collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden to Washington Headquarters Services, Directorate for Information Operations and Reports, 1215 Jefferson Davis Highway, Suite 1204, Arlington, VA 22202-4302, and to the Office of Management and Budget, Paperwork Reduction Project (0704-0188), Washington, DC 20503				
1. AGENCY USE ONLY (Leave blank)	2. REPORT DATE August 2004	3. REPORT TYPE AND DATES COVERED Final Jun 00 – Jun 04		
4. TITLE AND SUBTITLE PORTABLE OPEN SOURCE SECURITY ELEMENTS (POSSE)		5. FUNDING NUMBERS G - F30602-01-2-0537 PE - 62301F PR - CHAT TA - 00 WU - 05		
6. AUTHOR(S) (b) (6)				
7. PERFORMING ORGANIZATION NAME(S) AND ADDRESS(ES) University of Pennsylvania 200 South 33 rd Street Philadelphia PA 19104		8. PERFORMING ORGANIZATION REPORT NUMBER N/A		
9. SPONSORING / MONITORING AGENCY NAME(S) AND ADDRESS(ES) Defense Advanced Research Projects Agency 3701 North Fairfax Drive Arlington VA 22203-1714		10. SPONSORING / MONITORING AGENCY REPORT NUMBER AFRLIF-RS-TR-2004-230		
11. SUPPLEMENTARY NOTES AFRL Project Engineer: (b) (6)				
12a. DISTRIBUTION / AVAILABILITY STATEMENT DISTRIBUTION AUTHORIZED TO U.S. GOVERNMENT AGENCIES AND THEIR CONTRACTORS; CRITICAL TECHNOLOGY; AUG 04. OTHER REQUESTS FOR THIS DOCUMENT SHALL BE REFERRED TO AFRL/IFGA, ROME, NY 13441-4505.			12b. DISTRIBUTION CODE	
13. ABSTRACT (Maximum 200 Words) The most important contribution of POSSE is its stimulation of a security-conscious <i>community</i> of open source developers. <i>Portable Open Source Security Elements</i> (POSSE) is a means to accelerate realization of higher-assurance systems across <i>all open-source platforms</i> . Our approach is simple. We have constructed a consortium-like group of those focused on security from the open source operating systems community. We have the core groups of the OpenBSD operating system and the OpenSSL system for Apache (an open source web sever which is correctly viewed as an operating system). Our goal is to cross-fertilize amongst these core groups, develop security elements such as extended attributes for the Berkeley-derived Fast File System (FFS), and then export and evangelize these elements into the larger open source community, e.g., NetBSD and Linux. While we will discuss this in greater detail in the report, POSSE has <i>greatly accelerated</i> OpenBSD, Open SSL and Open SSH development efforts, and thus served to enhance the security of the e-commerce and information infrastructures.				
14. SUBJECT TERMS Open Source, Operating System, Open BSD, Open SSL			15. NUMBER OF PAGES 18	
			16. PRICE CODE	
17. SECURITY CLASSIFICATION OF REPORT UNCLASSIFIED	18. SECURITY CLASSIFICATION OF THIS PAGE UNCLASSIFIED	19. SECURITY CLASSIFICATION OF ABSTRACT UNCLASSIFIED	20. LIMITATION OF ABSTRACT UL	

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1. Introduction

Posse - *a group of people summoned by a sheriff to aid in law enforcement.*

Engineering a critical information infrastructure requires trustworthy software components. Such components have proven challenging to implement, as the work involved requires detailed analysis of the "fault-tree" of security penetrations, and coping with all the leaves of such a tree. The cost (and delay in time-to-market) for such analysis has inhibited commercial availability of such systems. Yet the need for high assurance systems grows with our reliance on the software components of our systems. While formal methods offer considerable promise, they have failed to successfully describe systems as deployed, and thus represent an additional cost beyond the already onerous analysis.

Our approach, *Portable Open Source Security Elements* (POSSE), has already made substantial progress. Our team is composed of the core teams of the OpenBSD, OpenSSH and OpenSSL projects with the addition of some University researchers. A key hypothesis we are exploring is that a *security-focused* UNIX variant can *inform* and *influence* the design of systems with similar ancestry and "raise the bar" for crackers. Any operating system monoculture, whether it be a version of Windows(TM) in the commercial sector or Linux in the open source sector, is extremely dangerous.

The most important contribution of POSSE is its stimulation of a security-conscious *community* of open source developers. *Portable Open Source Security Elements* (POSSE) is a means to accelerate realization of higher-assurance systems *across all open-source platforms*. Our approach is simple. We have constructed a consortium-like group of those focused on security from the open source operating systems community. We have the core groups of the OpenBSD operating system and the OpenSSL system for Apache (an open source web server which is correctly viewed as an operating system). Our goal is to cross-fertilize amongst these core groups, develop security elements such as extended attributes for the Berkeley-derived Fast File System (FFS), and then export and evangelize these elements into the larger open source community, e.g., NetBSD and Linux.

A second hypothesis is that such an approach, even if the social outreach fails, leaves DARPA with a higher-assurance choice (our workproduct, which will appear in at least the OpenBSD codebase), and is thus superior to funding a variety of technically interesting work for Linux with a much lower probability of entering the mainstream code base. In this way, the POSSE effort can be seen as a security spearhead, with important knock-on consequences for the open source community as a whole.

While we will discuss this in greater detail in the report, POSSE has *greatly accelerated* OpenBSD, OpenSSL and OpenSSH development efforts, and thus served to enhance the security of the e-commerce and information infrastructures.

2. Progress versus deliverables and SOW

Our major deliverables at the termination of the project were to be both a new version of OpenBSD available via the Web and on CD-Rom and demonstrated examples of security technology transitions among open source operating developers.

We have in fact delivered POSSE results *in five* releases of OpenBSD, OpenBSD 3.0 (released Fall 2001), OpenBSD 3.1 (released Spring 2002) and OpenBSD 3.2 (released Fall 2002). A *fourth* release was available via Web and CD-Rom effective May 1st, 2003, and OpenBSD 3.4 was available in Fall 2003, although management problems distorted the perceived value of the myriad technical contributions.

The detailed deliverables in the SOW were:

1. A series of enhancements to the OpenBSD operating system which will be in the mainstream distribution delivered on the CD-Rom and on the *openbsd.org* web site and its mirrors. These enhancements will include extended attributes for the BSD fast file system (FFS), following the model of TrustedBSD. They will similarly be integrated in informal collaboration with NAI Labs. We will work with (b) (6) to collaborate and cross-fertilize the projects and try to build a portable set of such enhancements. We will add ACLs for the filesystem built on top of the extended attributes.
2. The OpenBSD core team has extensive experience auditing the system. Many bugs have been found, years after the code has been in wide use. As part of the POSSE effort, we will apply this audit methodology to OpenSSL, logging and documenting the *methodology*, so that this process can be emulated in other system components. An additional positive consequence of this deliverable is that OpenSSL is widely used in e-commerce and other "infrastructure" contexts (ApacheSSL, *etc.*) which will have a broad impact on the assurance of systems where the SSL and TLS protocols are used. These audits are essential, as the cryptographic protocols and cryptographic algorithms are well-studied, but are subject to security failures as a consequence of poor/buggy implementations.
3. We will log and thoroughly document the auditing process. While we are not yet sure what form this documentation will take (on-line or off-line), there is considerable value in producing both. The idea we are currently pursuing, which leverages the strengths of the academic members of the team, is a book.
4. We will add support to OpenSSL and OpenSSH for symmetric-keyed encryption hardware. This will have a significant impact on performance, and thus increase usability. Note that the IPsec stack can already take advantage of encryption hardware.
5. We will add support for a */dev/policy* policy device to OpenBSD. The policy device provides a user-kernel interface for security/access-control policies. We have a rough prototype working for a distributed firewall system we have built under a DARPA-sponsored project (STRONGMAN) but it

needs to be generalized and audited for inclusion in a release. Besides the distributed firewall work, this will also be useful when porting some of the functionality introduced in the Security-enhanced Linux work.

6. Software systems will be delivered as operational software, distributed by both WWW and distribution CD-Roms. Updates will be available incrementally. Additional deliverables will include prototype applications used to test, audit and refine the system.

Our status on these items is as follows:

1. The enhanced file system has been designed and implemented by Dale Rahn in concert with Robert Watson of NAI Labs/TrustedBSD. The implementation effort has been kept completely synchronized with that of TrustedBSD.

2. Thorough and continuous auditing has been applied to OpenBSD and has resulted in the five releases of OpenBSD (3.0, 3.1, 3.2, 3.3 and 3.4) under the POSSE funding. The many bug fixes and modifications resulting from this effort (not all due to POSSE but impressive nevertheless) are detailed on the OpenBSD web site and the OpenSSL web site.

3. We have logged and documented the auditing process. While we are not yet sure what form this documentation will take (on-line or off-line), there is considerable value in producing both. While our intent was to produce a book, management challenges (see Section 4 below) have made this less likely as an option, although it remains an alternative if continued voluntarily or under other funding sources after project end.

4. We added support to OpenSSL and OpenSSH for symmetric-keyed encryption hardware, including a wide range of boards and chips. This is available in the releases; see Section 7 below.

5. The */dev/policy* policy device has been implemented for OpenBSD and continues being refined. As a major goal of this work was support for SE-Linux, we also undertook an effort (by Tom Langan of Penn) to provide SE-Linux features under OpenBSD. This was successful and is available, but not in the OpenBSD release. The */dev/policy* notions, including the use of advanced policy specification languages, were applied directly.

6. These software deliverables are available as OpenBSD 3.0, OpenBSD 3.1, OpenBSD 3.2, OpenBSD 3.3 and OpenBSD 3.4, on both CD-Rom and on the Internet, as well as OpenSSL on the OpenSSL web site.

In summary, we have met all of the proposed deliverables; an additional deliverable of integration with a Secure Bootstrap system being investigated by the University of Maryland was pushed into an option period and it does not appear that work will proceed. As can be seen from Section 5 below, we have exceeded what we proposed in many ways.

3. SOW Revision May 6, 2003

3.1 Original Statement of Work (Section E, verbatim from proposal)

Our major deliverables will include:

1. An audited OpenSSL release.
2. A portable kernel interface for OpenBSD with which OpenSSL can communicate with kernel-supported hardware cryptography devices.
3. An OpenSSL release with support for symmetric-keyed encryption hardware (such as 3DES) to increase performance.
4. File system enhancements to OpenBSD imported from TrustedBSD; ACLs built on top of the extended attributes from TrustedBSD.
5. Addition of a secure bootstrap component to OpenBSD, imported from the Maryland's SE-BOS.
6. Addition of the */dev/policy* device driver to the OpenBSD release, as a user-kernel interface for security policies.

All software developed will be made public ally available. There will be no patents associated with POSSE, and technical documentation will be made available through the World-Wide Web.

We view a major part of our work as outreach and developing a technical community of people capable of carrying out internal audit on their software. We will do that by sponsoring a total of four (4) POSSE "systems-fests" in Philadelphia at the University, where OpenBSD / OpenSSL developers can get together for 7-10 days in a laboratory environment. Members of the OpenBSD and OpenSSL communities would be supported by the substantial travel budget we have allocated. In addition, we would invite members of the Trusted FreeBSD and Secure Linux communities to be involved as well, both to enhance portability of the code and to build a community of security expertise. In essence, these would be "working workshops". They would be held every 6 months, probably 3 months offset from the semiannual CHATS PI meetings.

An additional outreach effort would be the attendance of POSSE project members at IETF, USENIX and FreeNIX meetings, especially security-related tracks.

3.2 New Statement of Work

Our major deliverables will include:

1. Continue the added work of the IHTFP letter proposal from (b) (6) to port the OpenBSD IPsec to Linux, working with the Linux community to get this into the mainstream.
2. Import performance improvements to the kernel cryptographic interface from FreeBSD (b) (6) (work) for OpenBSD with which, among other things, OpenSSL can communicate with kernel-supported hardware cryptography devices.
3. Add driver support for new hardware cryptography acceleration cards:
 - **netoctave**
 - **cavium**
 - **hifn**
 - **broadcom** (N.B., hifn and bcom have new variants out now or soon)
4. Study and characterize system performance enhancements due to hardware cryptography acceleration.
5. File system enhancements to OpenBSD imported from TrustedBSD; ACLs built on top of the extended attributes from TrustedBSD. These file system enhancements will be released either as a part of a future OpenBSD release or made available on a web site at the University of Pennsylvania, i.e., <http://www.cis.upenn.edu/dsl/POSSE/>. Specifically, ACL diffs from about 1 year ago (020523) were not fully reviewed; however the testing performed at the time had not found any problems. We need to apply these diffs to a -current tree, however it should be possible to merge these and obtain working ACL support. FreeBSD (where the code was derived from) has made some changes since this merger, it would be appropriate to bring in these changes.
6. Finish up remaining work on i386 $\widehat{W}$ support. Some of the initial assumptions we made when starting this are turning out poor, (e.g., 1G separation -it turns out that 256MB or 512 MB would work better and make more address space available to the programs).
7. PowerPC $\widehat{W}$ support, similar to the changes made on the i386 platform.
8. Further Apple machine support. It seems that they have made some changes to their latest generation of machines which requires some software changes. (Hardware is also required for this activity).

9. Advanced Power Management (APM) support on Apple laptop machines (specifically, suspend/sleep support).

10. Investigate PAE (36 bit physical memory addressing) support as it will require pmap changes.

11. Investigate Symmetric Multiprocessing (SMP) support on PowerPC, in concert with other work for the i386 platform (since most of the Apple desktop machines are now SMP, this addition would have considerable impact).

12. Fix ELF constructor/destructor code, which affects a few programs (wine), but needs to be corrected.

All software developed will be made publicly available¹ POSSE-developed software will continue to include the following copyright information:

```
*
* Redistribution and use in source and binary forms, with or
* without modification, are permitted provided that the following
* conditions are met:
*
* 1. Redistributions of source code must retain the above copyright *
notice, this list of conditions and the following disclaimer.
*
* 2. Redistributions in binary form must reproduce the above
* copyright notice, this list of conditions and the following
* disclaimer in the documentation and/or other materials provided
* with the distribution.
*
* 3. The name of the author may not be used to endorse or promote
* products derived from this software without specific prior
* written permission.
*
*
* THIS SOFTWARE IS PROVIDED BY THE AUTHOR "AS IS" AND ANY EXPRESS
* OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED
* WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE
* ARE DISCLAIMED. IN NO EVENT SHALL THE AUTHOR BE LIABLE FOR ANY
* DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL
* DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE
```

¹ as has been the case with the DARPA-supported 3.0, 3.1, 3.2 and 3.3 releases of OpenBSD

- * GOODS OR SERVICES; LOSS OF USE DATA, OR PROFITS; OR BUSINESS
- * INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY,
- * WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING
- * NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF
- * THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
- *
- * Effort sponsored in part by the Defense Advanced Research
- * Projects Agency (DARPA) and Air Force Research Laboratory, Air
- * Force Materiel Command, USAF, under agreement number
- * F30602-01-2-0537.

There will be no patents associated with POSSE, and technical documentation will be made available through the World-Wide Web.

We continue to view a major part of our work as outreach and developing a technical community of people capable of carrying out internal audit on their software. We intend to maintain the effort to work with TrustedBSD on the file system enhancement, continue to work with the OpenSSL project, continue to enhance Linux through the IPsec work, and continue to make bug fixes available as they are discovered.

Continued outreach will also be achieved via attendance of POSSE project members at IETF, SIGOPS, SIGCOMM, USENIX and FreeNIX meetings, especially security-related tracks.

4. Management and Discussion

We view a major part of our work as outreach and developing a technical community of people capable of carrying out internal audit on their software.

There have been a series of "hackfests" of about a week in duration held before or after major systems conferences. Among the products of this work is the new "pf" packet filter system and many other ideas and software systems. The basic architecture of the OpenSSL interface to symmetric-keyed cryptography accelerators was laid out at a hackfest. These meetings have been the genesis of new wireless security software and new methods of protecting operating systems from buffer overflow attacks. They have also served the need of community-building, and many of the results have been disseminated to other systems.

Management of the project, as anticipated, has been challenging. As we noted in the proposal there are the challenges of distributed management, strong personalities, and knitting together of sometimes quite

distinct development cultures. For example, the OpenSSL system must work across many operating systems and its collaboration is much looser - less structured, *etc.* than the OpenBSD development team, which is tightly integrated and led by Theo de Raadt. Thus, the OpenBSD model for what OpenSSL should look like is clear, while achieving larger scale consensus for OpenSSL took more time, leading to some tension. In the end, this was all smoothed by the many OpenSSL fixes, patches and enhancements that have emerged from both the OpenSSL auditing efforts and the OpenBSD cryptographic framework, cryptographic accelerator support, and OpenSSL support for OpenBSD.

One issue that did arise was that AL Digital's auditing efforts, which proved to be just "in the nick of time" (bug fixes just before a major exploit was published on the WWW), were expensive and AL Digital stopped work due to unavailability of funding until any further allocation from AFRL/DARPA arrives. A large body of auditing notes and the outline of a book on the OpenSSL audit has been produced but it is now unclear what the disposition of this information will be. Guidance would be appreciated, but an operating assumption is that some cleanup and publication on the WWW would extract the maximum value from these unique notes (where both the use of audit tools such as RATS and manual audit were performed).

A current status was provided to Air Force Research Laboratory dated April 15th, 2003 as part of the project reporting process. Budget and technical status were up to date and there is no point in reviewing such freshly provided material.

However, management of the project continued to prove challenging, in ways we had not anticipated. AFRL requested a work stoppage April 17th, 2003 for the project in its entirety, stimulating considerable media attention.

AFRL then revised the statement of work through memoranda on April 21st and April 24th to eliminate the "systems-fests" portion of the SOW and associated expenses such as subcontractors, and instructed the University of Pennsylvania to provide a revised statement of work in short order.

Our goal in the new statement of work was to take advantage of the fact that we have been more effective than expected in delivering security innovations and robust open source software. We have budgeted for the full amount of the original proposal with the "systems-fests" removed and some shifting of work from external contractors to the University. We have also added termination costs associated with the elimination of the "systems-fests" statement of work items, as specified in the letter of April 24th, 2003.

5. Selected Results

1. OpenSSL audit discovered and fixed holes in OpenSSL identified on the Internet. The holes in OpenSSL were fixed just before Defcon, and were totally due to CHATS funding. A patch of over 3,000

lines of code secures a host of problems of lesser severity, and generally hardens OpenSSL against future attacks,

(The OpenBSD crypto device support has also gone into OpenSSL and will be in the next release.)

2. Increased support of technologies for cryptography acceleration
3. Studied and characterized performance of cryptographic accelerators on network security protocols (Freenix paper)
4. OpenBSD Crypto Framework benchmarked and a paper submitted ((b) (6))
5. OpenBSD 3.2 distribution
6. ELF Id.so cleanup, macppc kernel profiling

Sparc [u]mul/[u]div/[u]rem optimization, If the processor supports the native multiply instruction allow dynamic executables to use them. This support was implemented in the dynamic loader.

W $\hat{X}$ (W xor X) code: most of the work was done in Nov/Dec of 2002, but final code checkin was 2003/01/17. This change greatly improves the execute permission situation with ELF executables and shared object. Without this change, the data region of the image was marked executable. This is due to the presence of the PLT in the DATA or BSS region of the image (true for alpha/sparc/sparc64/powerpc). The W $\hat{X}$ change causes multiple load section in the ELF image which allows each section to have the proper execute permissions.

The W $\hat{X}$ change also has one additional security feature: The GOT (Global Offset Table) is put into its own section to allow the dynamic linker to remove the write permissions of the GOT to prevent the running executable from modifying it. This prevents errant programs from modifying the GOT and causing other code to access the incorrect data/function.

-- commit message --

Module name: src

Changes by: ((b) (6)) 2003/01/17 13:50:13

Modified files:

```
gnu/usr.bin/binutils/bfd: elf.c
gnu/usr.bin/binutils/ld: genscripts.sh ld.h ldmain.c lexsup.c
gnu/usr.bin/binutils/ld/emulparams: elf32_sparc.sh elf32ppc.sh
                                   elf64_sparc.sh elf64alpha.s:
                                   elf_i386.sh
gnu/usr.bin/binutils/ld/scripttempl: elf.sc
```

Log message: This is a project to modify executables so that they do not have any executable regions which are writable. If a section of an executable is writable and executable, it is much easier for errant code to modify the executable's behavior.

Two current areas in shared library environments which have this critical problem are the GOT (Global Offset Table) and PLT (Procedure Linkage Table). The PLT is required to be executable and both GOT and PLT are writable on most architectures. On most ELF architecture machines this would cause shared libraries to have data and BSS marked as executable.

Padding to the linker script for programs and shared libraries/objects to isolate the GOT and PLT into their own load sections in the executables. This allows only the text (read only) region and the PLT region to be marked executable with the normal data and BSS not marked as executable. The PLT region is still marked executable on most architectures because the PLT lives in the "data" or "BSS" regions and the dynamic loader will need to modify it. Since the GOT and PLT should only ever be written by the dynamic linker, it will be modified to mprotect those regions so that they are not writable during normal execution. If the dynamic linker needs to modify the regions later, (*e.g.*, for lazy binding), it will mprotect the region, make the necessary changes, and mprotect it back. Since it is possible to receive a signal which would interrupt the program flow and perhaps cause the dynamic linker to modify the same (or nearby) PLT references, it is now necessary for signals to be blocked for the duration of the mprotect.

7. Significant blocks of new code were created as a result of the POSSE project. A current list, identified by the presence of the following addition to the OpenBSD copyright notice included in the source code:

```
* Effort sponsored in part by the Defense Advanced Research
* Projects Agency (DARPA) and Air Force Research Laboratory,
* Air Force Materiel Command, USAF, under agreement number
* F30602-01-2-0537.
*
```

is given below:

```
/usr/src/sys/arch/sparc64/dev/ce4231var.h
/usr/src/sys/arch/sparc64/dev/sab82532reg.h
/usr/src/sys/arch/sparc64/dev/uperfvar.h
/usr/src/sys/crypto/cryptodev.h
/usr/src/sys/dev/pci/hifn7751reg.h
/usr/src/sys/dev/pci/hifn7751var.h
/usr/src/sys/dev/pci/lofnreg.h
/usr/src/sys/dev/pci/lofnvar.h
/usr/src/sys/dev/pci/noctreg.h
/usr/src/sys/dev/pci/noctvar.h
```

```
/usr/src/sys/dev/pci/nofnreg.h
/usr/src/sys/dev/pci/nofnvar.h
/usr/src/sys/dev/pci/ubsecreg.h
/usr/src/sys/dev/pci/ubsecvar.h
/usr/src/sys/dev/sbus/asioereg.h
/usr/src/sys/dev/sbus/Cs4231var.h
/usr/src/sys/dev/sbus/spifreg.h
/usr/src/sys/dev/sbus/spifvar.h
/usr/src/sys/dev/sbus/uperf_sbuseg.h
/usr/src/sys/dev/sun/sunkbdreg.h
/usr/src/sys/dev/sun/sunkbdvar.h
/usr/src/sys/dev/sun/uperfio.h
/usr/src/sys/net/if_bridge.h
/usr/src/regress/lib/libc/atexit/atexit_test.c
/usr/src/sys/arch/powerpc/powerpc/pmap.c
/usr/src/sys/arch/sparc/dev/cgsix.c
/usr/src/sys/arch/sparc/dev/cgthree.c
/usr/src/sys/arch/sparc64/dev/beeper.c
/usr/src/sys/arch/sparc64/dev/Ce4231.c
/usr/src/sys/arch/sparc64/dev/comkbd_ebus.c
/usr/src/sys/arch/sparc64/dev/pckbc_ebus.c
/usr/src/sys/arch/sparc64/dev/sab.c
/usr/src/sys/arch/sparc64/dev/upa.c
/usr/src/sys/arch/sparc64/dev/uperf.c
/usr/src/sys/arch/sparc64/dev/uperf_ebus.c
/usr/src/sys/arch/sparc64/dev/vgafb.c
/usr/src/sys/crypto/cryptodev.c
/usr/src/sys/dev/pci/hifn7751.c
/usr/src/sys/dev/pci/lofn.c
/usr/src/sys/dev/pci/noct.c
/usr/src/sys/dev/pci/nofn.c
/usr/src/sys/dev/pci/ubsec.c
/usr/src/sys/dev/sbus/apio.c
/usr/src/sys/dev/sbus/asio.c
/usr/src/sys/dev/sbus/bwtwo.c
/usr/src/sys/dev/sbus/cgsix.c
/usr/src/sys/dev/sbus/cgthree.c
/usr/src/sys/dev/sbus/cs4231.c
/usr/src/sys/dev/sbus/spif.c
/usr/src/sys/dev/sbus/uperf_sbuse.c
```

```
/usr/src/sys/dev/sun/sunkbd.c  
/usr/src/sys/net/if_bridge.c  
/usr/src/sys/net/pf.c  
/usr/src/sys/net/pf_ioctl.c
```

This amounts to some 36768 lines of code in OpenBSD 3.2.

8. Evaluated the RNG ofubsec and hifn 7951 with a FIPS-140-2 test suite that is hooked into the RNG subsystem (this work isn't committed yet, it needs a little more time to percolate): as a result, moved ubsec to using the SHA1'd version of its RNG output instead of the "raw" results (it wouldn't pass FIPS-140-2 very often in raw mode). Hifn needs the ANSI DES extension to pass the test, and doesn't provide that as easily in hardware as Broadcom.

9. Several fixes in the bridge and 802.1 D spanning tree protocol code

10. Fix information leak in ViaTech Rhine driver (ethernet padding bug)

11. New implementation of tcpdump/print-gre.c (old version had license problems)

12. Sun Crypto 1000 now supported (It's a rebadged Broadcom 5821)

13. Added CRYPTO_NULL xform so we can measure API overhead.

14. ubsec/lofn/hifn all enabled on i386, macppc, and sparc64

15. Added simplistic load balancing of crypto sessions, simplified registration ((b) (6))

16. "The Design of the OpenBSD Cryptographic Framework" ((b) (6)) will appear in USENIX'03 (June 2003).

17. "Performance Analysis of Cryptography in Network Security Protocols" ((b) (6)) Work in Progress, 2003.

18. Propolice integration: worked with etoh to fix about 50 major bugs in it. Propolice is now a very solid software product, and etoh is building up enough credit over it that it now has potential for gcc integration.

19. Began investigating randomized shared object layout, and perhaps even randomized main program layout.

20. In the process of a string audit of the entire tree. First, change all string handlings to bounded truncations. Next, going to start on truncation handling - if something is cut, make a decision.

21. Continued to modify setuid and daemon programs so that they jail themselves more, and better. Additional priv separation and priv reduction went into the tree.

22. The 3.3 and 3.4 releases of OpenBSD.